

E-Commerce Sales Analytics — MySQL

An end-to-end SQL project built on a custom-designed e-commerce database representing a Pakistani retail business. The database covers customers, products, orders, employees, and regional sales data across five cities.

The project demonstrates the full range of SQL skills — from basic filtering and aggregation to joins across multiple tables, subqueries, CASE WHEN logic, views, and stored procedures. It also covers advanced topics including window functions (RANK, LAG, NTILE, running totals), recursive CTEs for org chart traversal, indexes for query optimization, and DML operations for real-world data maintenance.

The capstone query combines CTEs, window functions, and conditional logic to rank every customer by lifetime value, assign city-level rankings, and label them into spending tiers — Platinum through Bronze.

Tools: MySQL | **Concepts:** Joins, Aggregations, Subqueries, Views, Window Functions, CTEs, Stored Procedures, Indexes, DML

```

14 -- Q2. All orders that were delivered in 2023
15 • SELECT
16     order_id,
17     customer_id,
18     order_date,
19     shipped_date,
20     status
21 FROM orders
22 WHERE status = 'Delivered'
23     AND order_date BETWEEN '2023-01-01' AND '2023-12-31'
24 ORDER BY order_date;
25
26

```

Result Grid   Filter Rows: Edit:    Export/Import:   Wrap Cell Content: 

	order_id	customer_id	order_date	shipped_date	status
▶	1	1	2023-01-15 10:00:00	2023-01-17 14:00:00	Delivered
	2	2	2023-02-20 11:30:00	2023-02-22 09:00:00	Delivered
	3	3	2023-03-05 09:15:00	2023-03-07 11:00:00	Delivered
	4	4	2023-04-10 14:00:00	2023-04-12 10:00:00	Delivered
	6	6	2023-06-22 08:45:00	2023-06-24 15:00:00	Delivered
	7	7	2023-07-30 12:00:00	2023-08-02 10:00:00	Delivered
	8	8	2023-08-14 10:30:00	2023-08-16 13:00:00	Delivered
	10	10	2023-10-05 11:00:00	2023-10-07 16:00:00	Delivered
	11	1	2023-11-11 13:45:00	2023-11-13 11:00:00	Delivered
▲	NULL	NULL	NULL	NULL	NULL

orders 2 x

Output

 Action Output ▼

#	Time	Action	Message
1	12:39:51	SELECT order_id, customer_id, order_date, shipped_date, status FROM orders WHERE status = 'Delivered' AND ord...	9 row(s) returned

```

56 -- Q6. Calculate the revenue for each line item in an order
57 • SELECT
58     order_id,
59     product_id,
60     quantity,
61     unit_price,
62     discount_pct,
63     ROUND(quantity * unit_price * (1 - discount_pct), 2) AS line_revenue
64 FROM order_items
65 ORDER BY order_id;

```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 

	order_id	product_id	quantity	unit_price	discount_pct	line_revenue
1	1	1	1	85000.00	0.00	85000.00
1	3	1	1	8500.00	0.05	8075.00
2	2	1	1	195000.00	0.10	175500.00
3	10	2	2	1800.00	0.00	3600.00
3	11	1	1	950.00	0.00	950.00
4	4	3	3	3200.00	0.05	9120.00
4	5	2	2	4500.00	0.05	8550.00
5	13	1	1	6200.00	0.00	6200.00
6	7	1	1	5500.00	0.00	5500.00
6	8	1	1	9200.00	0.10	8280.00
7	12	2	2	3500.00	0.00	7000.00
7	14	1	1	12000.00	0.00	12000.00
8	9	2	2	2800.00	0.00	5600.00
9	15	1	1	45000.00	0.08	41400.00
10	6	2	2	7800.00	0.05	14820.00
11	1	1	1	85000.00	0.00	85000.00

result 5 x

Output

 Action Output 

#	Time	Action	Message
1	12:48:47	SELECT order_id, product_id, quantity, unit_price, discount_pct, ROUND(quantity * unit_price * (1 - discount_pct), 2...	21 row(s) returned

```
193 -- Q18. Full order breakdown: customer, product, category, and line total
194 • SELECT
195     o.order_id,
196     CONCAT(cu.first_name, ' ', cu.last_name) AS customer,
197     p.product_name,
198     cat.category_name,
199     oi.quantity,
200     oi.unit_price,
201     oi.discount_pct,
202     ROUND(oi.quantity * oi.unit_price * (1 - oi.discount_pct), 2) AS line_total
203 FROM orders o
204 JOIN customers cu ON o.customer_id = cu.customer_id
205 JOIN order_items oi ON o.order_id = oi.order_id
206 JOIN products p ON oi.product_id = p.product_id
207 JOIN categories cat ON p.category_id = cat.category_id
208 ORDER BY o.order_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	order_id	customer	product_name	category_name	quantity	unit_price	discount_pct	line_total
▶	1	Ali Raza	Samsung Galaxy S24	Electronics	1	85000.00	0.00	85000.00
	1	Ali Raza	Wireless Earbuds	Electronics	1	8500.00	0.05	8075.00
	2	Sana Noor	Laptop Dell XPS 13	Electronics	1	195000.00	0.10	175500.00
	3	Junaid Ahmed	Data Science Book	Books	2	1800.00	0.00	3600.00
	3	Junaid Ahmed	Urdu Novel Set	Books	1	950.00	0.00	950.00
	4	Ayesha Butt	Men's Kurta Set	Clothing	3	3200.00	0.05	9120.00
	4	Ayesha Butt	Women's Lawn Suit	Clothing	2	4500.00	0.05	8550.00
	5	Tarin Mahmood	Cricket Bat Kashmir	Sports	1	6200.00	0.00	6200.00

Result 13 x

Output

Action Output ▼

#	Time	Action	Message
✓ 1	13:07:50	SELECT o.order_id, CONCAT(cu.first_name, ' ', cu.last_name) AS customer, p.product_name, cat.category_name, oi.q...	21 row(s) returned


```

87
88 -- Q9. How many customers do we have in each segment?
89 • SELECT
90     segment,
91     COUNT(*) AS total_customers
92 FROM customers
93 GROUP BY segment
94 ORDER BY total_customers DESC;
95

```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 

	segment	total_customers
▶	Retail	5
	VIP	3
	Wholesale	2

Result 8 x

Output

 Action Output ▼

	#	Time	Action	Message
✓	1	12:54:03	SELECT segment, COUNT(*) AS total_customers FROM customers GROUP BY segment ORDER BY total_customers DESC ...	3 row(s) returned

```

145  |-- Q14. Every order with the customer name and the employee who handled it
146  •  SELECT
147      o.order_id,
148      CONCAT(c.first_name, ' ', c.last_name) AS customer_name,
149      c.city,
150      CONCAT(e.first_name, ' ', e.last_name) AS handled_by,
151      o.order_date,
152      o.status
153  FROM orders o
154  INNER JOIN customers c ON o.customer_id = c.customer_id
155  INNER JOIN employees e ON o.employee_id = e.employee_id
156  ORDER BY o.order_date;

```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 

	order_id	customer_name	city	handled_by	order_date	status
▶	1	Ali Raza	Karachi	Zainab Malik	2023-01-15 10:00:00	Delivered
	2	Sana Noor	Lahore	Bilal Hussain	2023-02-20 11:30:00	Delivered
	3	Junaid Ahmed	Islamabad	Nadia Sheikh	2023-03-05 09:15:00	Delivered
	4	Ayesha Butt	Karachi	Zainab Malik	2023-04-10 14:00:00	Delivered
	5	Tariq Mehmood	Peshawar	Hassan Raza	2023-05-18 16:30:00	Cancelled
	6	Hira Qureshi	Lahore	Bilal Hussain	2023-06-22 08:45:00	Delivered
	7	Usman Ghani	Quetta	Hassan Raza	2023-07-30 12:00:00	Delivered
	8	Mariam Iqbal	Karachi	Zainab Malik	2023-08-14 10:30:00	Delivered
	9	Kamran Baig	Islamabad	Nadia Sheikh	2023-09-01 09:00:00	Processing
	10	Zara Waheed	Lahore	Bilal Hussain	2023-10-05 11:00:00	Delivered

Result 11 

Output



Action Output



	#	Time	Action	Message
✓	1	13:04:10	SELECT o.order_id, CONCAT(c.first_name, ' ', c.last_name) AS customer_name, c.city, CONCAT(e.first_name, ' ', e.last_...	15 row(s) returned

```

158
159 -- Q15. All customers and how many orders each one has placed
160 • SELECT
161     c.customer_id,
162     CONCAT(c.first_name, ' ', c.last_name) AS customer_name,
163     c.segment,
164     COUNT(o.order_id) AS total_orders
165 FROM customers c
166 LEFT JOIN orders o ON c.customer_id = o.customer_id
167 GROUP BY c.customer_id, customer_name, c.segment
168 ORDER BY total_orders DESC;

```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 

	customer_id	customer_name	segment	total_orders
▶	1	Ali Raza	Retail	2
	2	Sana Noor	VIP	2
	3	Junaid Ahmed	Wholesale	2
	5	Tariq Mehmood	Retail	2
	6	Hira Qureshi	VIP	2
	4	Ayesha Butt	Retail	1
	7	Usman Ghani	Wholesale	1
	8	Mariam Iqbal	Retail	1
	9	Kamran Baig	VIP	1
	10	Zara Waheed	Retail	1

Result 12 x

Output

 Action Output ▼

#	Time	Action	Message
✓ 1	13:06:03	SELECT c.customer_id, CONCAT(c.first_name, ' ', c.last_name) AS customer_name, c.segment, COUNT(o.order_id) AS t...	10 row(s) returned

```
45
46 -- Q5. Orders that haven't been shipped yet
47 • SELECT
48     order_id,
49     customer_id,
50     order_date,
51     status
52 FROM orders
53 WHERE shipped_date IS NULL;
54
```

Result Grid

  Filter Rows:

Edit:   

Export/Import:  

Wrap Cell Content: 

	order_id	customer_id	order_date	status
▶	5	5	2023-05-18 16:30:00	Cancelled
	9	9	2023-09-01 09:00:00	Processing
	12	2	2023-12-01 10:15:00	Shipped
	15	6	2024-03-20 11:00:00	Pending
•	NULL	NULL	NULL	NULL

orders 4 x

Output

 Action Output ▼

#	Time	Action	Message
1	12:46:13	SELECT order_id, customer_id, order_date, status FROM orders WHERE shipped_date IS NULL LIMIT 0, 1000	4 row(s) returned


```
26
27  |-- Q3. Products priced between 5,000 and 20,000 (cheapest first)
28  •  SELECT
29      product_name,
30      unit_price,
31      stock_qty
32  FROM products
33  WHERE unit_price BETWEEN 5000 AND 20000
34  ORDER BY unit_price ASC;
--
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	product_name	unit_price	stock_qty
▶	Rice Cooker 1.8L	5500.00	60
	Cricket Bat Kashmir	6200.00	75
	Running Shoes	7800.00	90
	Wireless Earbuds	8500.00	150
	Non-stick Cookware	9200.00	40
	Dumbbell Set 10kg	12000.00	35

products 3 x

Output

Action Output ▼

#	Time	Action	Message
✓ 1	12:41:58	SELECT product_name, unit_price, stock_qty FROM products WHERE unit_price BETWEEN 5000 AND 20000 ORDER B...	6 row(s) returned

```

213
214 -- Q19. Customers who have spent more than the average order value
215 • SELECT
216     cu.customer_id,
217     CONCAT(cu.first_name, ' ', cu.last_name) AS customer_name,
218     ROUND(SUM(oi.quantity * oi.unit_price * (1 - oi.discount_pct)), 2) AS total_spent
219 FROM customers cu
220 JOIN orders o ON cu.customer_id = o.customer_id
221 JOIN order_items oi ON o.order_id = oi.order_id
222 WHERE o.status != 'Cancelled'
223 GROUP BY cu.customer_id, customer_name
224 HAVING total_spent > (
225     SELECT AVG(order_total)
226     FROM (
227         SELECT SUM(quantity * unit_price * (1 - discount_pct)) AS order_total
228         FROM order_items
229         GROUP BY order_id
230     ) AS sub
231 )
232 ORDER BY total_spent DESC;

```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	customer_id	customer_name	total_spent
▶	2	Sana Noor	341250.00
	1	Ali Raza	193375.00

```

96
97  -- Q10. Total revenue generated by each order
98  •  SELECT
99      oi.order_id,
100      ROUND(SUM(oi.quantity * oi.unit_price * (1 - oi.discount_pct)), 2) AS order_revenue
101  FROM order_items oi
102  GROUP BY oi.order_id
103  ORDER BY order_revenue DESC;
104

```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 

	order_id	order_revenue
▶	2	175500.00
	12	165750.00
	11	100300.00
	1	93075.00
	9	41400.00
	7	19000.00
	4	17670.00
	10	14820.00
	6	13780.00
	14	12400.00
	5	6200.00
	8	5600.00
	13	5400.00
	3	4550.00

Result 9 

Output



Action Output

#	Time	Action	Message
✓ 1	12:56:21	SELECT oi.order_id, ROUND(SUM(oi.quantity * oi.unit_price * (1 - oi.discount_pct)), 2) AS order_revenue FROM order_items o...	15 row(s) returned

```

66
67
68 -- Q7. Find all products that have "Set" in their name
69 • SELECT
70     product_name,
71     unit_price,
72     stock_qty
73 FROM products
74 WHERE product_name LIKE '%Set%';
75

```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 

	product_name	unit_price	stock_qty
▶	Men's Kurta Set	3200.00	200
	Urdu Novel Set	950.00	250
	Dumbbell Set 10kg	12000.00	35

products 7 x

Output

 Action Output ▼

	#	Time	Action	Message
✓	1	12:50:37	SELECT product_name, unit_price, stock_qty FROM products WHERE product_name LIKE "%Set%" LIMIT 0, 1000	3 row(s) returned